



Assessment of Finnish Power-to-X Economy research 2017–2023: Scientific articles and research activities

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ABSTRACT

Power-to-X technologies are considered as one of the most viable options for solving the energy and climate challenges of our time. First, significant greenhouse gas reductions are needed to reach the 1.5°C climate target of the Paris Agreement in the course of this century. Second, recent geopolitical threats have highlighted the risks associated with energy and fossil fuel dependence. In addition, sustainable technologies, including Power-to-X, are expected to bring significant investments, economic growth, and jobs to Finland, which is why it is one of the most discussed energy topics in society. However, the development is not in mature stages when it comes to technological readiness, let alone the implementation of related technologies. This study conducts a comprehensive overview of Power-to-X research in Finland applying bibliometric methods and survey, through which the current knowledge on the topic is revealed, how it is distributed among different institutions and researchers, and what are the current strengths and research gaps. The study seeks to promote Power-to-X research and the related green transition, not only in Finland, but throughout the European Union, aiming at being climate neutral by 2050.

1. Introduction

The accumulation of greenhouse gases, such as carbon dioxide (CO₂), in the Earth's atmosphere is driving climate change, which represents one of the most dangerous impacts on our environment resulting from current energy production practices [1]. Each ton of CO₂ emitted into the atmosphere exacerbates global warming, whereas avoiding these emissions helps slow the rate of temperature increase [2]. This principle is central to European Climate Law, which sets the ambitious goal for the European Union (EU) to achieve net-zero GHG emissions by 2050 [3]. To achieve this objective, the EU is focusing on reducing emissions through investments and advancements in green technology, as well as the preservation of natural resources. A notable example of this ambition is the recent allocation of a 720 M€ fund to support renewable hydrogen production. When completed, it is expected for the initiative to have the potential to prevent 10 megatons (Mt) of CO₂ emissions over a 10-year period [4]. Among 27 EU member states, Finland stands out for its ambitious climate goals, aiming for net-zero carbon emissions by 2035 [5]. However, EEA's (The European Environment Agency) recent report indicates that EU member states GHG reduction falls short (49% drop by 2030) of EU's 2030 target of 55%. Significant emission reductions are still achievable, but they require sustained efforts to reach the

2030 targets and beyond. Here, the expansion of renewable electricity production and improvement of energy efficiency would be effective methods for reducing GHG emissions and reaching EU targets in 2030 and beyond [6]. A major challenge to a successful energy transition is the political commitment and determination of Member States. Opposition to a more sustainable society often arises from certain sectors and individuals, with social unrest and strong resistance coming from public opinion [7].

Renewable energy sources (RES) are pivotal in transition towards sustainability, anticipated to provide the requisite power for directly electrifying various sectors, including industrial processes, transportation, and components of district heating and cooling systems [8–11]. Research for the energy transition in Finland revealed the need for RES and a comprehensive electrification of the energy system [12–18]. Direct electrification emerges as a promising avenue for reducing CO₂ emissions, particularly due to the notable decrease in the carbon intensity of energy driven by the widespread adoption of RES [19–21].

CO₂ emissions from electricity production in Finland have decreased 83% between 2010 and 2023. In Finland, a substantial proportion of the national electricity consumption of 78 TWh in 2023 was CO₂-free, with 54% derived from RES [22]. Especially the share of wind power in the RES has risen radically: in 2010 the wind power produced 0.79

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TWh electricity and in 2023 this had increased to 14.13 TWh of electricity supply in Finland [22]. However, as RES integration escalates in electrical grids, ensuring power supply reliability becomes progressively challenging due to the dynamic and variable nature of RES generation [23,24]. Moreover, specific industries encounter challenges in direct electrification due to the yet to be solved technical complexities associated with current technology, such as battery limitations to heavy-duty transport [24–26]. Similarly, decarbonizing the chemical industry poses distinctive challenges given its heavy reliance on carbon sources for large-volume chemical production [27–29]. These complexities of complete industrial defossilisation highlight the multifaceted nature of transitioning towards sustainable practices in various sectors, underscoring the need for innovative solutions and strategic intervention.

Introducing "Power-to-X" (PtX) technologies [30] provides a potential solution to indirectly electrify complex sectors [24,31,32]. PtX technologies enable the storage of electrical energy in chemicals, allowing for flexibility in energy use and distribution [23,33]. The first step in the PtX system pathway, known as "Power-to-hydrogen," involves the production of renewable hydrogen [34–36]. This hydrogen can then be used directly as an industrial feedstock or mobility fuel, or further processed with nitrogen, oxygen, or carbon to produce chemical compounds such as ammonia, methanol, methane, and kerosene jet fuel and derivative fuels, which can serve as alternatives to conventional fossil fuels and chemical feedstocks [37–40]. Prior life-cycle assessment studies indicate the PtX solutions outperform fossil fuel-based products by their sustainability, namely by reducing GHG emissions [41–43]. These solutions include PtX in plastic production [41], protein and ammonia production [43] and steelmaking [42]. Finland possesses many of the necessary components to foster the development of a PtX economy [37], whether through favorable natural resources or supportive regional and national policies. However, recent academic investigations into the PtX transition in Europe indicate that the Nordics still lag behind major European industrial players. For instance, [44], reviewed the status of PtX project demonstrations in Europe and concluded that significant variability exists in the development stage of processes among countries. Notably, Germany and France emerged prominently in terms of the number of commissioned and planned projects, as well as capacity. Similarly, Decourt [45] carried out a technological innovation system study and found that while the PtX system is building in Europe, this is developing unevenly and since the 2010s, and project development has largely been led by Germany, France, and the UK.

The gap between Finland and other European nations in PtX project development prompted this study. As a nascent field, the successful development of a viable PtX value chain requires optimization in process pathways, energy demand modeling, system coupling, and more, which are areas typically explored in academic institutions such as universities and research centers [46,47]. Therefore, to comprehensively understand the current status, potential future directions, and limitations of the Finnish PtX system, *this study conducted the first comprehensive overview of the current status of the entire Finnish Power-to-X economy.* The analysis comprised two main steps. First, a bibliometric analysis of articles and scientific publications with corresponding authorship in Finnish research organizations between 2017 and 2023 was conducted. Bibliometric analyses employ quantitative approaches to article metadata, enabling statistical analysis to evaluate the significance of topics within a research field and understand the impact of specific publications, authors, and institutions [48]. Overall, bibliometric studies offer a systematic approach to discerning research trends and gaps within a given topic. Secondly, to gain insights into the research infrastructure facilitating PtX research in Finland and to identify future research aims, information about the research facilities and strategies of twelve participating Finnish research centers was obtained through interviews.

Overall, this study aims to examine the development of PtX related topics, the distribution of research among organizations, research teams, and authors. Additionally, the research investigates the terminology used in PtX research, organizational focus, infrastructure, and activity

of participating research institutions from Finland. These findings are analyzed in terms of annual development, impact, collaboration, and mutual influence, offering a starting point to observe Finnish PtX research and its development and facilitating international comparisons. Importantly, it is hoped that the findings of this study will promote the research and understanding of PtX economy and encourage similar and comparative studies in other parts of the world. Moreover, the results can be used in the development of national research strategies globally and to increase the domestic and international collaboration in research related to the PtX economy.

2. Methodology and research objectives

This study conducted a comprehensive analysis on the status of PtX research in Finland between 2017 and 2023. Here, a bibliometric analysis of the relevant literature was employed and interviews and surveys from research organizations were assessed according to their most prominent research topics and infrastructure within the PtX research.

The objective of the bibliometric analysis was to identify all the articles aligned with the research framework published within the period of interest. These articles were then checked against a novel framework and articles that aligned with it were included in this study and consequently classified according to their topic and subtopic. Bibliometric analysis techniques were then applied to the selected articles to elucidate the distribution of research within the PtX research in Finland, at the individual and research group levels. Bibliometric studies are a suitable method for systematically analyzing large datasets within broad research areas, as is the case with the PtX topic [49]. Within this study, the chosen bibliometric approach was science mapping [49], as the primary aim of this work was to analyze the Finnish PtX research from three following perspectives:

- i. Identify the key topics researched:
 - a. Distribution of topics
 - b. Annual development
 - c. Impact
- ii. Identify the research distribution of organizations, teams, and authors:
 - a. Topic distribution within organizations
 - b. Teams involved
 - c. Authors involved
- iii. Identify the key collaborations:
 - a. Domestic and international

The second part of this study was conducted with surveys and follow-up interviews with participating institutions. The twelve included Finnish research institutions were the Aalto University, LUT University, Natural Resources Institute Finland (LUKE), Tampere University, Technical Research Centre of Finland (VTT), University of Eastern Finland, University of Helsinki, University of Jyväskylä, University of Oulu, University of Turku, University of Vaasa, and Åbo Akademi University. The interview and survey elements of the study were key not only for creating the development of the research infrastructure and framework used here, but also for gaining a better understanding about the most prominent publications, research groups, focus areas, and infrastructure within each university related to PtX research in Finland. In addition, interviews were also used to develop the framework of this study, providing the opportunity for experts to provide feedback.

2.1. Survey and interview data collection

The data collection from the participating organizations started with a survey about the status of their PtX research, to which all twelve research organizations responded. This survey element included three discrete sections. The first section of the survey enquired about published articles within the PtX concept. The aim of this section was to obtain complementary articles to include in the posterior data analysis phase

of this study to ensure that all the articles deemed relevant by the surveyed institutions were considered in the present study.

The second section of the survey focused on gaining an understanding about the test infrastructure related to PtX research to identify existing resources and facilities available within Finnish universities for conducting PtX research. In this section, institutions were asked to give the following information about their testing facilities:

- i. Organization
- ii. Infrastructure Name
- iii. Equipment Name
- iv. Description
- v. Research Focus
- vi. Infrastructure Size
- vii. Access Policy
- viii. User Fee
- ix. Technology Readiness Level
- x. Contact Person

The final section of the survey explored the research teams and general focus of the institution within PtX research. In the final section, all participating institutions were requested input data regarding:

- i. Organization
- ii. Faculty
- iii. Team leader
- iv. Team member
- v. Degree Position
- vi. Research focus of each individual

After the surveys, interviews were conducted to all the Finnish research organizations, except the University of Helsinki to validate the results and discuss the emerging framework. Additionally, in these interviews the validity and inclusion of keywords influenced by the developing framework were presented to the institutions listed in section two.

2.1.1. Framework

The interviews with research organizations enabled to highlight all the important aspects of the PtX Economy concept, which made it possible to create the framework of this study (see Fig. 1) and identify the keywords (see Appendix 1) for the literature search. The research framework (Fig. 1) was divided into three main sections: Sustainable Energy Transformation, Regulatory and Material Aspects, and Analytical Perspectives. Within the Sustainable Energy Transformation, eight main

categories (A) have been identified, including Power-to-X (PtX) Economy, CO₂ Capture, Renewable Hydrogen Production, Carbon Dioxide, Hydrogen, Other By-products in Hydrogen Production, PtX technologies, and Applications. These categories were further divided (Appendix 2) into more detailed subcategories B and C, which are more detailed categories for the topics. The regulatory and material aspect was dedicated to exploring the regulatory landscape, raw materials, and materials essential for sustainable energy projects. Analytical perspectives focused on computational modeling, techno-economic analysis, data analysis, and process optimization. The final step in the framework creation was to align it with Statistics Finland's Science 2010 Classification [50]. The framework was utilized as the inclusion criterion in the bibliometric search and in the categorization of the focus areas of researchers, research teams and faculties across all participating Finnish institutions and as the main tool for the bibliometric analysis.

2.2. Bibliometric study: status of Power-to-X research in Finland

2.2.1. Bibliometric methodology description

The first step in the bibliometric analysis was to identify all the literature linked to the field of PtX published in peer reviewed scientific articles by Finnish research institutions between 2017 and 2023 by a Scopus search. In this study, a total of 97 keywords and keyword combinations (see Appendix 1) were used within the Scopus database search. Additionally, articles that the research organizations identified in the interviews were included in the bibliometric analysis.

The Scopus search was conducted with the above-mentioned keywords and keywords combinations, and the search was conducted within the article title, abstract, keywords. The search included all the Finnish universities and research institutions included in section two, as well as the University of Lapland, and Hanken School of Economics. The Scopus search included all the above-mentioned relevant Finnish research institutions, and these were included in each search within the affiliation name of the Scopus search alongside with the keywords and keyword combinations. Quotation marks were applied between each keyword and research affiliation. The Scopus search was conducted between 16 and 17th of October 2023. The following criteria within the keyword search were set for the inclusion of the articles: i) Data range: 2017–2023, ii) Document type: Peer-reviewed scientific article, iii) Source type: Journal, iv) Publication stage: Final (published), v) Language: English.

Additionally, key exclusions to the results were included in the Scopus search. Firstly, articles that belonged to the subject areas of

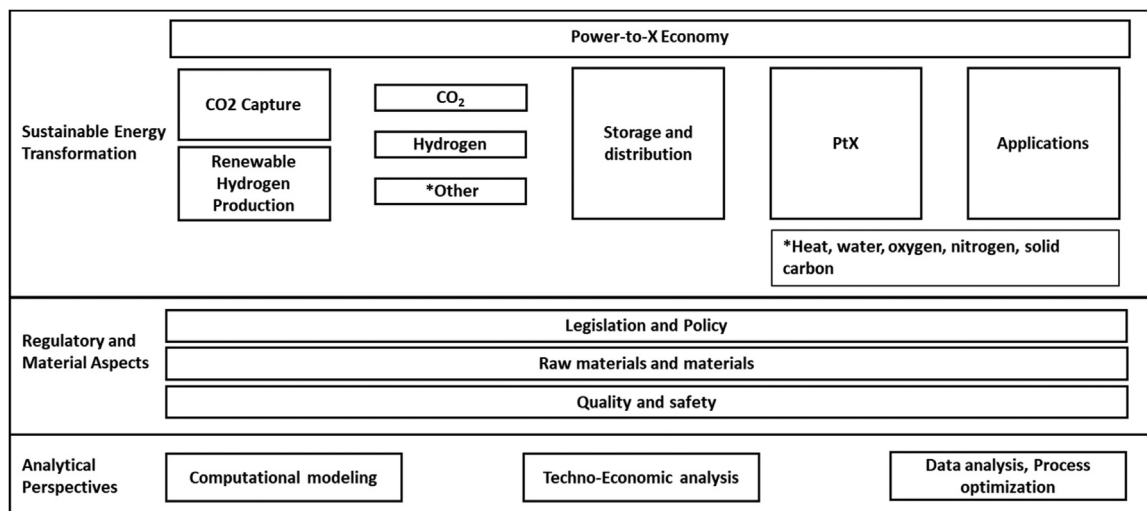


Fig. 1. Framework for PtX Economy Research (fuels, chemicals, materials).

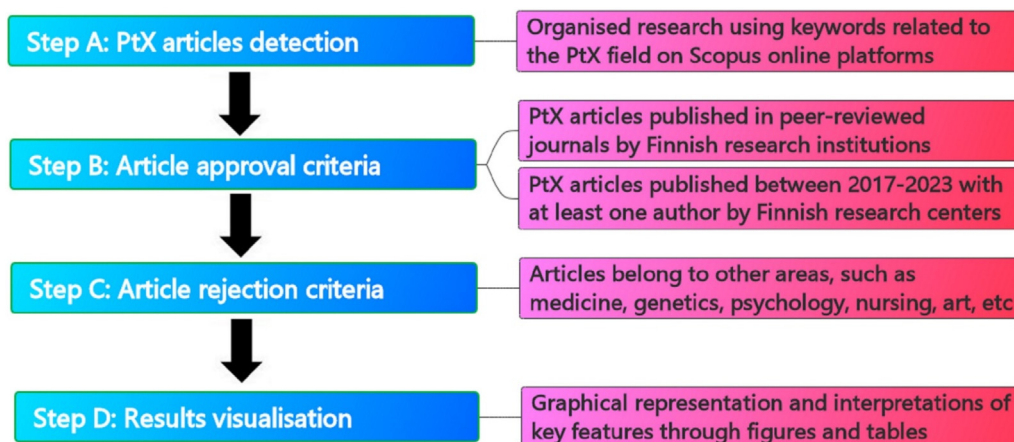


Fig. 2. The methodological scheme details the procedure of the applied method, including the database, criteria, output data, and visual diagrams.

medicine, biochemistry, genetics and molecular biology, earth and planetary sciences, neuroscience, immunology and microbiology, pharmacology, toxicology and pharmaceuticals, psychology, nursing, dentistry, health professions, arts and humanities, or veterinary were intentionally excluded. Finally, it was ensured that at least one of the corresponding authors was affiliated with a Finnish research institution. Fig. 2 outlines the procedures employed for the above-mentioned benchmarks.

2.2.2. Bibliometric data analysis

The analysis of the Scopus search results began by extracting the Scopus list of all included articles (Scopus and articles provided by research institutions) into a csv-file for MS Excel, with the following information: authors, title, year, source title (publication), cited by, link (of publication in Scopus), affiliations, authors with affiliation, abstract, author keywords, index keywords, correspondence address.

Consecutively, articles were classified and categorized in MS Excel file with the following information: publication year, authors, corresponding authors, corresponding authors in Finland, corresponding organizations in Finland, international collaboration institutions, countries (linked to international collaboration), domestic collaboration institutions, industry collaboration, title, source, link, citation count, abstract, author keywords, index keywords, framework categories (A-H), and article type (original research paper or review).

The last step of the Scopus search data analysis involved the individual classification of each article in the framework categories (A-H) based on the detail included in the abstracts and index words. Overall, the selection yielded 334 articles (295 from Scopus search and 39 from research institutions).

2.2.3. Bibliometric study: Power-to-X concept term incidence

To evaluate the incidence of the main terms used in academic literature to refer to the PtX concept, a new term search was performed in Scopus. This analysis was conducted on a global scale using the terms “Power-to-X”, “PtX” and “P2X” directly as queries within article title, abstract, keywords. Additionally, when searching for “PtX” and “P2X” the search was refined by including the term “hydrogen” as an additional query term to limit the incidence of articles unrelated to the PtX term framework of this study. In the next phase an inclusion criterion similar as outlined in Section 2.2.1, considering subject area limitations and article requirements, were applied to this search. However, the data range was adjusted to cover the period from 2010 to 2023. The term incidence analysis was performed on February 2nd, 2024. The third phase consisted of the article inclusion analysis, which was primarily based on its title and relevance to the study framework outlined in Section 2.2.

Overall, this analysis yielded 1581 articles related to the PtX concept (from “Power-to-X”, “PtX”, and “P2X”).

3. Results

3.1. Power-to-X alternative terms analysis

Once the relevant exclusion parameters were applied to the raw data, 398 articles were found in the Scopus term search for “Power-to-X”, “PtX” and “P2X”. Of the three terms, the one that had the highest overall incidence between 2010 and 2023 was “Power-to-X” (293), followed by “PtX” (71) and “P2X” (34), as shown in Fig. 3.

Fig. 3 showed the yearly incidence of term usage relating to the PtX concept. Moreover, this showed that up until 2017 there were less than two articles published every year discussing the ‘Power-to-X’, ‘PtX’ or ‘P2X’ concept. However, the general trends of term usage began to change from 2017 onwards as this year marked the first time within the data range where more than two articles were published using the same term, in the same year (Fig. 2) (“Power-to-X” = 3; “PtX” = 5; “P2X” = 1). After 2017 the usage of “Power-to-X”, “PtX” and “P2X” began to diverge, with the utilization of “Power-to-X” notably increasing relative to other terms. By 2020, “Power-to-X” had become the most frequently used term, while “P2X” term was barely utilized (“Power-to-X” = 52; “PtX” = 11; “P2X” = 1). Subsequently, since 2020, the utilization of “Power-to-X” nearly doubled by 2023, while the annual utilization of “PtX” or “P2X” did not exceed 17 articles during the same time period (“Power-to-X” = 96; “PtX” = 17; “P2X” = 13) (Fig. 2).

Moreover, the analysis revealed that most of the articles found in the “PtX” and “P2X” category were unrelated to the PtX concept as defined in this article. In the “PtX” results, 83% of all articles were found to be out of the framework as they investigated applications of the chemotherapeutic agent Paclitaxel, or PtX, within the medical field (2074 out of 2475 articles). Similarly, 82% of all the “P2X” articles belonged to the field medicine and mostly investigated the P2X- cellular receptor (190 out of 232 articles). The terminology of all not relevant disciplines for the scope of this study, such as medicine, was excluded from the analysis.

3.2. Article analysis

The bibliometric analysis yielded a total of 334 articles (see Table 1) and the distribution of articles in Finnish research organizations are displayed in Fig. 4. In sections below, it is analyzed how the findings of these articles are distributed to the applied framework and what type of research have been utilized within them.

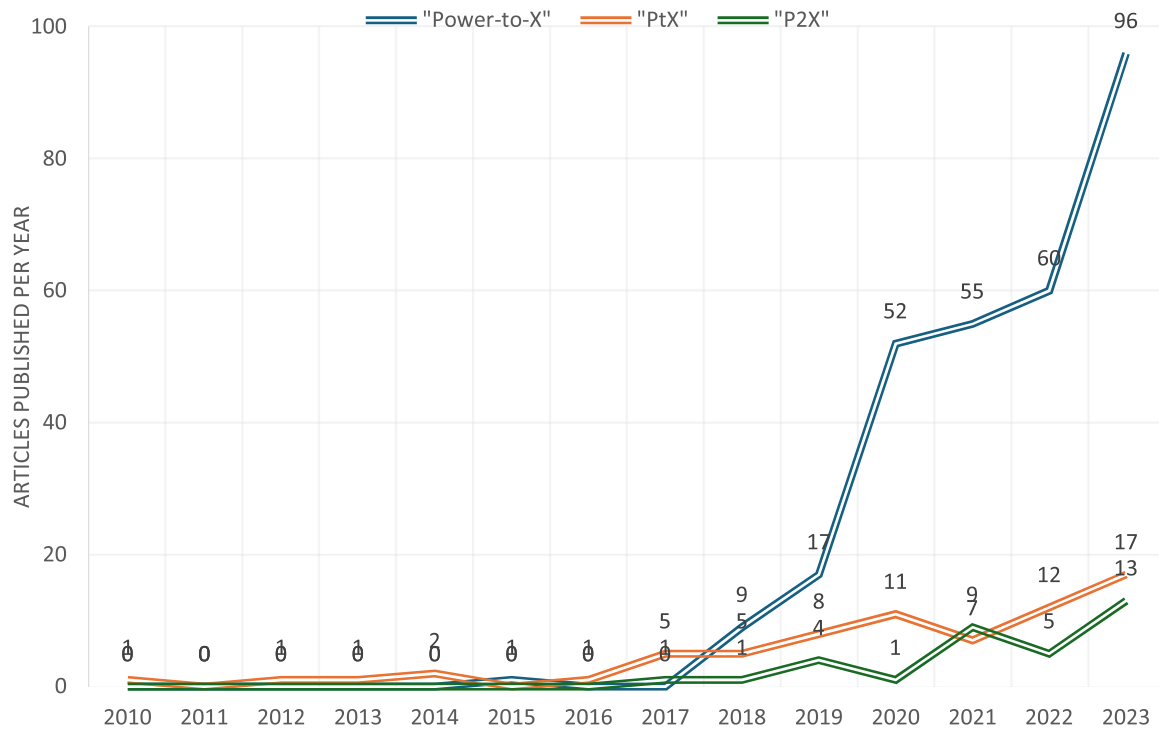


Fig. 3. Yearly incidence of PtX related term between 2010 and 2023.

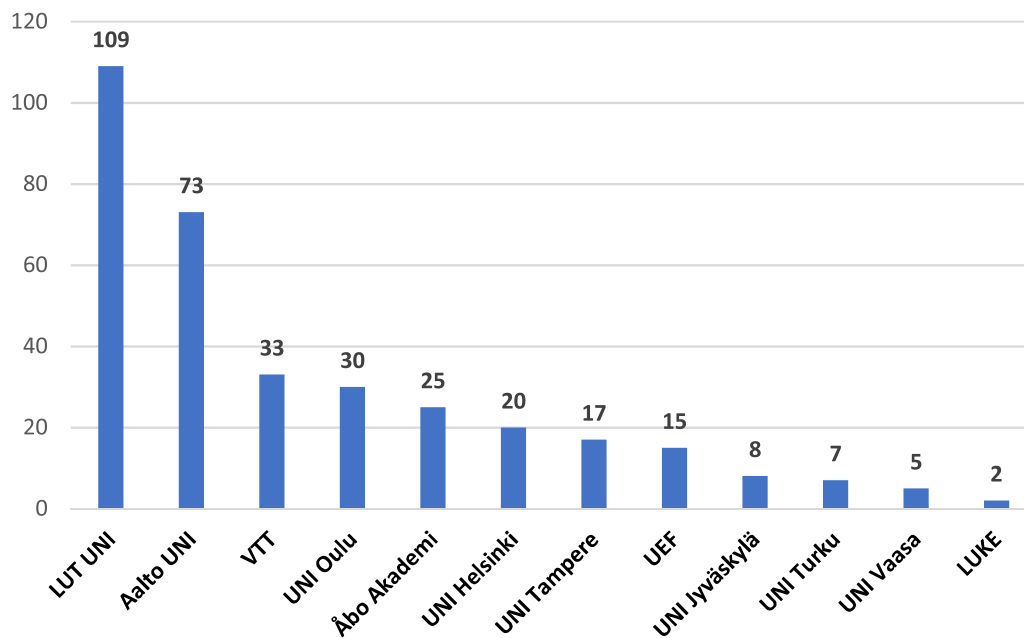


Fig. 4. Article distribution in Finnish research organizations.

Table 1
Number of articles and annual distribution.

2017	2018	2019	2020	2021	2022	2023	TOTAL
28	31	33	57	49	67	69	334

3.2.1. Article distribution by topics and research types

The analysis started with investigation how the findings from the dataset distributed to the framework of this study. The findings of this analysis are presented below by categories. The analysis also illustrates

the distribution of material and raw material, legislation and policy and safety related findings and research types.

The analysis of category A (Fig. 5) illustrates that PtX economy is the most researched topic, which counts 25.2% of the findings in the dataset. The next most addressed topics are renewable hydrogen production (16.7%), CO₂ capture (16.4%), hydrogen (12.3%), and PtX technologies (11.8%). These are followed by carbon dioxide (7.1%), applications (5.4%), and other by-products in hydrogen production (5.1%). More detailed results of these in the B and C categories (see Appendices 3 and 4) are presented below.

The majority of the **PtX economy** related findings link to energy system integration and concepts, which counts 21% from the entire findings

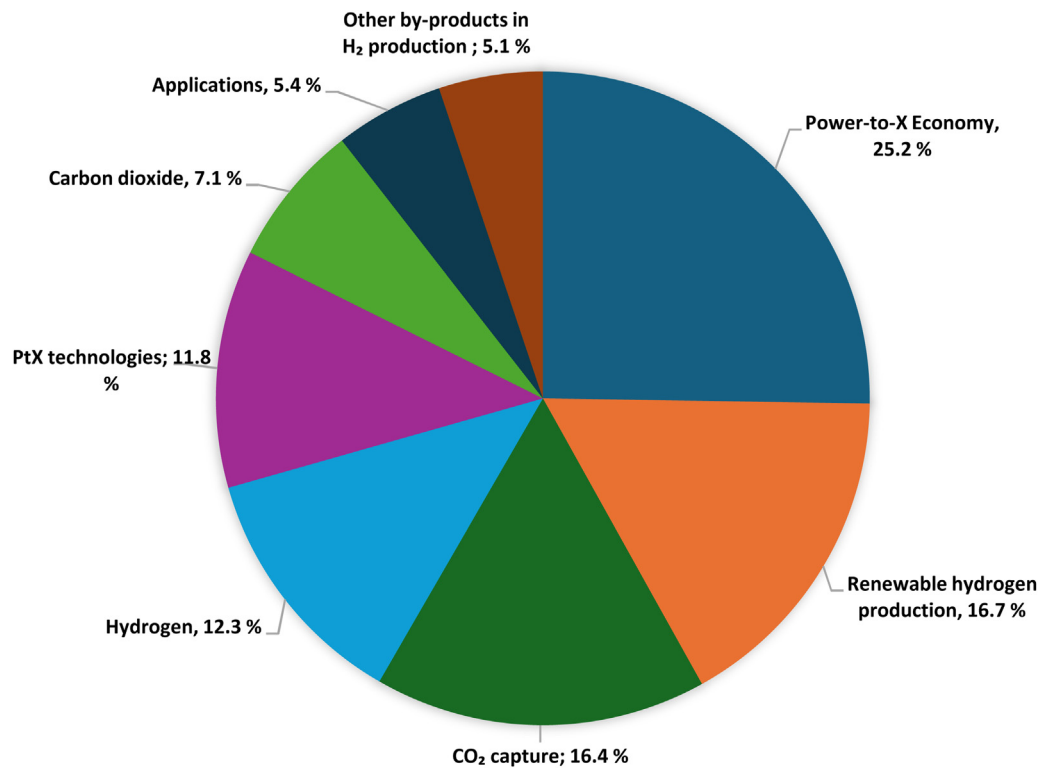


Fig. 5. Distribution of findings in category A.

in category B. The more in-depth investigation of category C illustrates that energy system research (31.7%) is the most represented topic within the energy system and integration. The analysis reveals that other topics have received limited or no research at all. In addition, the analysis indicates that environmental and sustainability aspects (9.6%) are much less represented in the findings than energy system integration and concepts. The findings in this category are mainly focused on the climate change (14.4%), leaving other topics in a minor role or has not been studied at all.

Renewable hydrogen production related findings are divided into a bundle of four: electrolysis (7.0%), photocatalysis (4.3%), and photoelectrochemical (3.4%) hydrogen production. Here, the analysis illustrates that the remaining topics in categories B and C had a very minor role or had not been studied at all. The findings of **CO₂ capture** are mostly linked to industry sources (7.5%), leaving Direct Air Capture (DAC) (1.4%) in a clear minority. The further analysis shows that CO₂ capture related findings are very limited in the category C. **Hydrogen** related findings were distributed among all its uses. Nevertheless, the findings were most focused on the hydrogen storage (5.8%), utilization (4.3%) and distribution (3.6%), leaving hydrogen embrittlement (1.9%) in a clear minority in the dataset. The analysis of category C illustrated that the research is limited. Analysis of the **PtX technologies** related findings illustrates that it leans towards Power-to-Liquids (PtL) (6.7%) and Power-to-Gas (PtG) (3.9%), leaving Power-to-Material (PtM) (1.0%) in a clear minority. Moreover, the analysis of category C shows that only e-methanol and derived products (6.0%) has received some attention. **Carbon dioxide** related findings were focused on the utilization (5.1%) aspect, where storage (1.7%) had very limited findings and distribution none. The findings related to **applications** seem to be limited in general, where only sustainable steel (2.7%) had some research. Findings on **other by-products in hydrogen production** were entirely linked to the heat (5.1%), with no other topics represented in the dataset. The analysis shows that heat related findings had limited representation in the category C.

The analysis on the proportion of **materials and raw materials**, legislation and policy and safety aspects within the dataset illustrated the following highlights. First, materials and raw materials oriented focused research articles represented 8.4% of the entire dataset (28 articles out of 334). Here, the analysis illustrated that majority of the findings were linked to renewable hydrogen production (53.6%) in the category A. These were further distributed in the category B to electrolysis (30.4%), photocatalysis (17.4%), and photoelectrochemical hydrogen production (8.7%). Second most studied topic in materials and raw materials was hydrogen, representing 32.1% from the higher (category A) findings. The findings on this topic related to hydrogen embrittlement (21.7%), hydrogen storage (13.0%) and hydrogen distribution (8.7%) in the category B. Thirdly, CO₂ capture related findings represented 14.3% of the findings in the higher category A but did not break down into more detailed subjects.

The **legislation and policy** related articles represented 6.9% of the dataset (23 articles out of 334). The findings of these articles distributed into legislation (47.4%) and policy instrument research (42.1%). The analysis of category B illustrated that the findings in legislation and regulation, and policy instruments and subsidies were dominated by the PtX economy topic. However, the number of findings in each was limited and that was very little or no research at all in more detailed categories. The analysis on the **quality and safety** aspects related findings revealed that these were practically not researched at all.

The analysis on **research types** showed that there were 14 review articles on the dataset, counting 4.2% of the dataset. The distribution of findings in review articles was highly fragmented and limited to make conclusions. Moreover, the analysis revealed that there were 28 techno-economic analyses utilizing articles (8.4%) in the dataset. The majority of techno-economic related findings were linked to PtX technologies (27.9%) and PtX economy (20.9%) topics in the category A and (PtL) (21.4%) and energy system integration and concepts (19.0%) in category B. The distribution in category C was too fragmented and limited to make significant insights.

Table 2
Overall citation development.

	2017	2018	2019	2020	2021	2022	2023
Number of articles	28	31	33	57	49	67	69
Average citations	36.75	44.23	47.18	23.91	27.10	13.39	1.94
Total citations	1029	1371	1557	1363	1328	897	134

Table 3
Citation development in category A.

A CATEGORY	2017	2018	2019	2020	2021	2022	2023	TOTAL
Power-to-X Economy	91	499	360	356	691	295	66	2358
CO ₂ capture	272	167	795	339	110	292	25	2000
Renewable hydrogen production	471	411	160	222	199	211	24	1698
PtX	200	273	117	99	304	158	25	1176
Hydrogen	43	33	57	255	102	30	29	549
Carbon dioxide	16	28	131	44	60	186	18	483
Applications	0	98	160	126	34	26	16	460
Other	75	59	40	24	68	44	8	318

The key takeaways from this section:

- PtX Economy leads by the volume of findings, especially linked to energy system research.
- Environmental and sustainability issues were also represented, namely linked to climate change.
- Renewable hydrogen production (electrolysis), CO₂ capture (industry), hydrogen (storage), and PtX technologies (PtL, e-methanol and derived products), carbon dioxide (utilization), and other by-products in H₂ production (heat) topics stood out in the number of findings.
- By number of findings, there are several topics that warrant more research and research gaps that have not been studied at all.

3.2.2. Citation development

The annual citation development is illustrated in [Table 2](#), indicating that the most cited year is 2019. However, the citation development in recent years cannot be estimated based on the number of citations, but this is analyzed in more detail in [Section 3.2.8](#).

The topic analysis in category A ([Table 3](#)) illustrates that PtX economy is the most cited topic, followed by CO₂ capture, renewable hydrogen production and PtX technologies with over 1000 citations each. The rest of the citations distribute to hydrogen, carbon dioxide, applications and other by-products in hydrogen production. The analysis on annual citation development illustrates that renewable hydrogen production had most of the citations at the beginning of the observation period in 2017 and CO₂ capture was the most cited topic in 2019. Otherwise, PtX economy has been the most cited topic in 2018 and between 2020 and 2023.

The citation analysis in categories B and C (see Appendices 5 and 6) included only topics with 100 or more citations. This analysis illustrated that energy system integration and concepts and environmental

and sustainability aspects linked to Power-to-X Economy represented the most cited topics in the category B. Other impactful topics, reaching 500 or more citations, in the category B were (renewable hydrogen production) electrolysis, (CO₂ capture) DAC and industry, and (PtX technologies) PtG. The annual citation analysis showed that the (renewable hydrogen production) electrolysis topic was the most cited in 2017 and (CO₂ capture) DAC in 2019. Otherwise, the most cited topic was energy system integration and concepts during the observation period.

The analysis on citation development in category C shows that the energy system topic is the most cited topic within the framework of this study by total citations and has been the most cited topic for five years out of the seven-year observation period 2017–2023. Moreover, climate change represents an impactful topic within this category. The rest of topics that had 100 or more citations are (PtG) e-ammonia, (CO₂ capture) industry – biogenic CO₂, (heat) utilization, (PtM) e-food, (PtL) e-methanol and derived products, (environmental and sustainability aspects) social acceptance, (heat) storage, (hydrogen utilization) electricity, and (sustainable traffic system) maritime.

The key takeaways from this section:

- The energy system research has gained most of the citations overall and annually. Climate change has also gained a decent provision of the citations. These insights reflect the overall volume of findings.
- Other cited topics included CO₂ capture (industry – biogenic CO₂, DAC), renewable hydrogen production (electrolysis, photocatalysis, photoelectrochemical), PtX (PtG – e-ammonia, PtL – e-methanol and derived products, PtM – e-food), hydrogen (storage, utilization – electricity), carbon dioxide (utilization), applications (sustainable traffic system – maritime), and other by-products of renewable hydrogen production (heat – utilization and storage).
 - When comparing these with the number of findings, it seems that CO₂ capture (industry – biogenic CO₂, DAC), renewable hydrogen production (photocatalysis, photoelectrochemical), PtX (PtG – e-ammonia, PtM – e-food), hydrogen (utilization), applications (sustainable traffic system – maritime and other by-products in renewable hydrogen production (heat – utilization and storage) have relatively high citations as all these topics represent limited numbers in volume.

3.2.3. Distribution of findings in Finnish research organizations

The analysis on how the findings distribute by research organizations was done through average analysis. Here, the objective was to highlight in which topics different organizations had above the average number of findings, which enabled to pinpoint the research orientation of different organizations.

The category A analysis (see [Table 4](#)) revealed that Aalto University and LUT University illustrate the most comprehensive orientation in most of the topics. Aalto University has above the average number of findings in all topics except carbon dioxide. LUT University has above the average amount of findings in all except other by-products in hydrogen production. In addition, the analysis illustrated that VTT has the above the average amount of findings in PtX technologies, hydrogen, and other by-products in hydrogen production. Åbo Akademi

Table 4
Occurrence of A-category findings in Finnish research organizations.

	Power-to-X Economy	Renewable hydrogen production	CO ₂ capture	Hydrogen
Leading	LUT UNI	Aalto UNI	LUT UNI	LUT UNI
Average or above	Aalto UNI UNI Helsinki	Åbo Akademi UNI UNI Oulu Tampere UNI LUT UNI	Aalto UNI Åbo Akademi UNI	Aalto UNI UNI Oulu VTT
Leading	PtX technologies LUT UNI	Carbon dioxide LUT UNI	Applications LUT UNI	Other by-products in hydrogen production Aalto UNI
Average or above	Aalto UNI VTT	N/A	Aalto UNI UNI Oulu	Åbo Akademi UNI VTT

University's research is highlighted in renewable hydrogen production, CO₂ capture, and other by-products in hydrogen production. University of Oulu's findings stand out in renewable hydrogen production, CO₂ capture and carbon dioxide. Moreover, the University of Helsinki had above the average occurrence of findings in Power-to-X Economy, while UNI Tampere had it on renewable hydrogen production.

The analysis on category B (see Appendix 7) was conducted so that only topics that had ten or more findings in total were included. This analysis revealed that the LUT University, University of Oulu, Aalto University, Tampere University, VTT, and Åbo Akademi University were the leading organizations in at least one of the category B topics. Moreover, the analysis revealed that most of the research organizations had above the average representations in several category B topics. The analysis of category C (see Appendix 8) included only topics that had five or more findings represented in total. This analysis illustrates that the number of topics were drastically reduced in more detailed research. Here, the LUT University, Aalto University, VTT, University Jyväskylä and University of Helsinki were organization that had leading positions at least in one of the category C topics. Moreover, the Åbo Akademi University and University of Eastern Finland had topics where they had above the average number of findings.

The analysis also included the investigation of how material and raw material research distributed within the Finnish research organizations. Here, the findings indicate that the leading material and raw material-oriented organization was Aalto University (10 occurrences in findings), followed by the University of Turku (4) and VTT (4). In addition, the analysis revealed that the material research in Aalto University was focused on renewable hydrogen production, especially on electrolysis and material research on hydrogen was linked to hydrogen embrittlement. The material research in the University of Turku was focused on renewable hydrogen production, more precisely on photocatalysis. VTT's material research was steered towards hydrogen, but findings were fragmented to single hits on embrittlement, storage and distribution.

The analysis on legislation and policy related findings in the dataset indicates that this is mainly focused on two research organizations

Table 5
Author development.

2017	2018	2019	2020	2021	2022	2023
97	123	111	193	217	335	262

in Finland, the University of Eastern Finland (UEF) and University of Helsinki. Here, the analysis illustrates that University of Eastern Finland's research is mostly linked to legislation over policy issues. Moreover, most of the legislation related findings in University of Eastern Finland were linked to Power-to-X Economy, more precisely in energy system integration and concepts and energy system-oriented research. The findings of the University of Helsinki remained in the higher category A, where the subject area was mostly linked to the Power-to-X Economy in general.

The key takeaways from this section:

- There is a strong research focus from the majority of Finnish research organizations towards the PtX topics and that nearly all organizations have at least one topic where they excel.
- In addition, the analysis revealed that there are certain organizations focused on material and raw material, and legislation and policy research.
- However, findings indicate that there are several topics that have not been addressed in Finnish research organizations.

3.2.4. The number of authors and distribution of research in journals

The total number of authors in the dataset according to VOSviewer was 930. The overall trend indicates a growth in involved researchers (see Table 5), as the number of authors has increased every year except 2023. The year 2023 is not fully comparable, but the number of authors was already nearly two and a half times more in October 2023 than at the start of the observation period 2017.

The distribution of the dataset in different journals are presented in Fig. 6, which covers the 16 most published journals. Most of the research

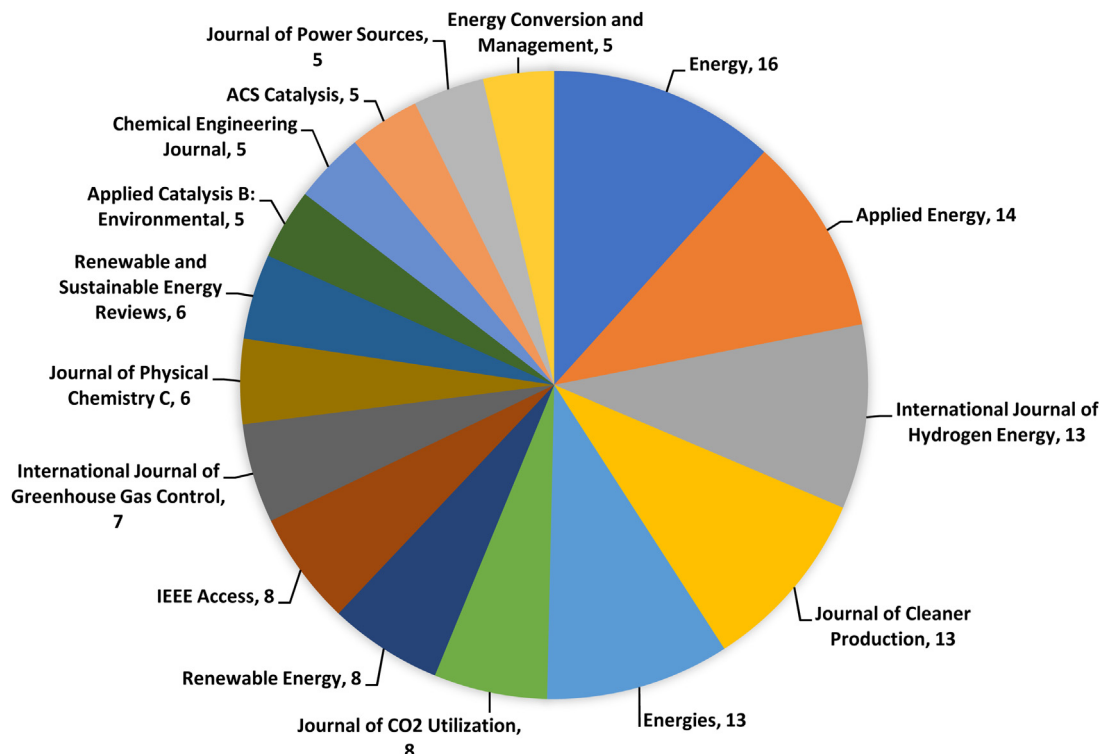


Fig. 6. Distribution of Finnish Power-to-X Economy research in top 16 journals.

Table 6
Domestic and industry collaboration.

Industry collaboration	Corresponding organization	Aalto UNI	LUKE	LUT UNI	Tampere UNI	UEF	UNI Helsinki	UNI Jyväskylä	UNI Oulu	UNI Turku	UNI Vaasa	VTT	Åbo Akademi	TOTAL
7	Aalto UNI	–	1	1	1	0	1	1	2	1	0	3	0	11
0	LUKE	0	–	0	0	0	2	0	0	1	0	0	0	3
9	LUT UNI	1	0	–	1	1	0	0	3	0	0	8	2	16
0	Tampere UNI	0	0	0	–	1	2	1	0	1	0	1	0	6
1	UEF	0	0	0	1	–	4	0	1	1	0	0	0	7
1	UNI Helsinki	2	2	0	0	3	–	0	0	1	0	1	0	9
0	UNI Jyväskylä	0	0	0	0	0	0	–	0	0	0	0	0	0
10	UNI Oulu	6	0	1	0	0	0	0	–	0	0	1	0	8
0	UNI Turku	1	0	0	0	0	0	0	0	–	0	0	0	1
0	UNI Vaasa	0	0	0	0	0	0	0	0	0	–	0	0	0
8	VTT	4	0	1	1	0	2	0	0	0	0	–	1	9
4	Åbo Akademi	1	0	0	0	0	0	0	1	3	0	0	–	5

articles were published in Energy, Applied Energy, International Journal of Hydrogen Energy, Journal of Cleaner Production, and Energies journals, which had all more than 10 articles each.

The citation distribution in journals is visible in Appendix 9. This table included only journals that reached 120 or more citations in total. The analysis shows that the Journal of Cleaner Production is clearly the most cited journal within the dataset, followed by Applied Energy, Energy, ACS Catalysis, International Journal of Hydrogen Energy, and the Chemical Engineering Journal, which have all 300 or more citations in total.

The key takeaways from this section:

- Overall, there is a strong correspondence of researchers working in the PtX topic in Finland and the number seems to be growing.

3.2.5. Collaboration

The analysis of domestic and international collaboration in Finnish research organizations was conducted by listing articles that included domestic or international collaboration in separate Excel sheets. The domestic collaboration yielded a total of 63 articles, which corresponds 18.9% of the dataset. A more detailed analysis (see Table 6) illustrates the collaboration between Finnish research organizations. In terms of numbers, most of the collaboration has been made done between LUT University and VTT (8), followed by collaboration between the University of Oulu and Aalto University (6). Moreover, VTT and Aalto University, University of Eastern Finland and University of Helsinki have more cooperation than average. Overall, the analysis illustrates that domestic collaboration is limited and many organizations have not collaborated extensively during the observation period of the study in PtX economy related research in Finland. The analysis of industry collaboration (Table 6) revealed that there were 40 articles involving collaboration with industry companies, representing 12% of the dataset. Here, the University of Oulu, LUT University, VTT, Aalto University, and Åbo Akademi University represent the most active organizations conducting industry collaboration within the dataset.

The analysis of international collaboration revealed that there were 124 articles, which had cooperation extending outside of Finland, representing 37.1% of the dataset. This analysis illustrated that 61.4% of the international collaboration in Finnish PtX research is conducted within Europe. The collaboration with the rest of regions is as follows: Asia (19.6%), North America (6.5%), Middle East (4.3%), Latin America (3.8%), Africa (2.2%), Oceania (1.1%), and Eurasia (1.1%). The country level analysis (see Appendix 10) illustrates that the top five countries of international collaboration are China (23), Sweden (19), Germany (16), Italy (13), and France (9). Other prominent cooperation countries with Finnish PtX research are the Netherlands, UK, USA, Norway, Ireland, Iran, Canada, Denmark, Spain, Austria, and Belgium, which all represent at least four articles or more. Analysis on the international collaboration institutions (see Appendix 11), illustrates that the most prominent

international collaboration organizations are by volume Umeå University (9), Tianjin University (6), Chalmers University of Technology (5), Tulane University (4), Xiangtan University (4), and Hainan University (4).

The key takeaways from this section:

- The domestic collaboration in Finnish PtX research is limited.
- The international collaboration is active, mainly within Europe, but China is the most popular location by country.
- Industry collaboration is focused on certain organizations in Finnish PtX research.

3.2.6. Research teams

The research teams were identified in the following manner. First, a VOSviewer tool was utilized for csv-file from Scopus, which included all the articles in the dataset. After this, a map-based analysis was conducted with VOSviewer on co-authorship, where the unit of analysis was authors. This applied a full counting method, included a maximum of 25 authors per document, included authors with the minimum of three documents and set no minimum number of citations. This procedure yielded 137 authors.

The research group analysis was conducted in four phases. The first phase included only the items that were related to each other. This yielded a total of 10 clusters. The second phase analysis included those items that were not connected to each other, which yielded 20 clusters. The third phase of analysis compared the clusters from the first two steps in Excel. The criterion for a research group was that there had to be at least two members in a research group, a connection between the team members to each other, the same organization with the team members, and that the team had to represent a Finnish research organization. In the fourth phase, the articles on the dataset were grouped according to the research groups. This yielded 20 research teams (Fig. 7 and Table 7) from the dataset. The research group analysis illustrates that majority of the research groups originate from LUT University (6) and Aalto University (5), which represent together more than half of the research teams discovered. Moreover, the University of Oulu (3) and VTT (2) had more than a single research group. The leading research groups by overall citation were Breyer et al. (RG 1), Laasonen et al. (RG 3), Kiviluoma et al. (RG 4), and Ahola et al. (RG 2) having more than 400 citations each.

The analysis of collaboration between the top research groups (Table 8) shows that research group 16 (Honkala et al., UNI Jyväskylä) and research group 12 (Santasalo-Aarnio et al., Aalto UNI) and research group 3 (Laasonen et al., Aalto UNI) have the most active research collaboration within the dataset.

The key takeaways from this section:

- The analysis illustrated some collaboration between top research groups in Finnish PtX research.
- Noticeable is also that there are several research groups that do not illustrate any collaboration within the discovered research groups.

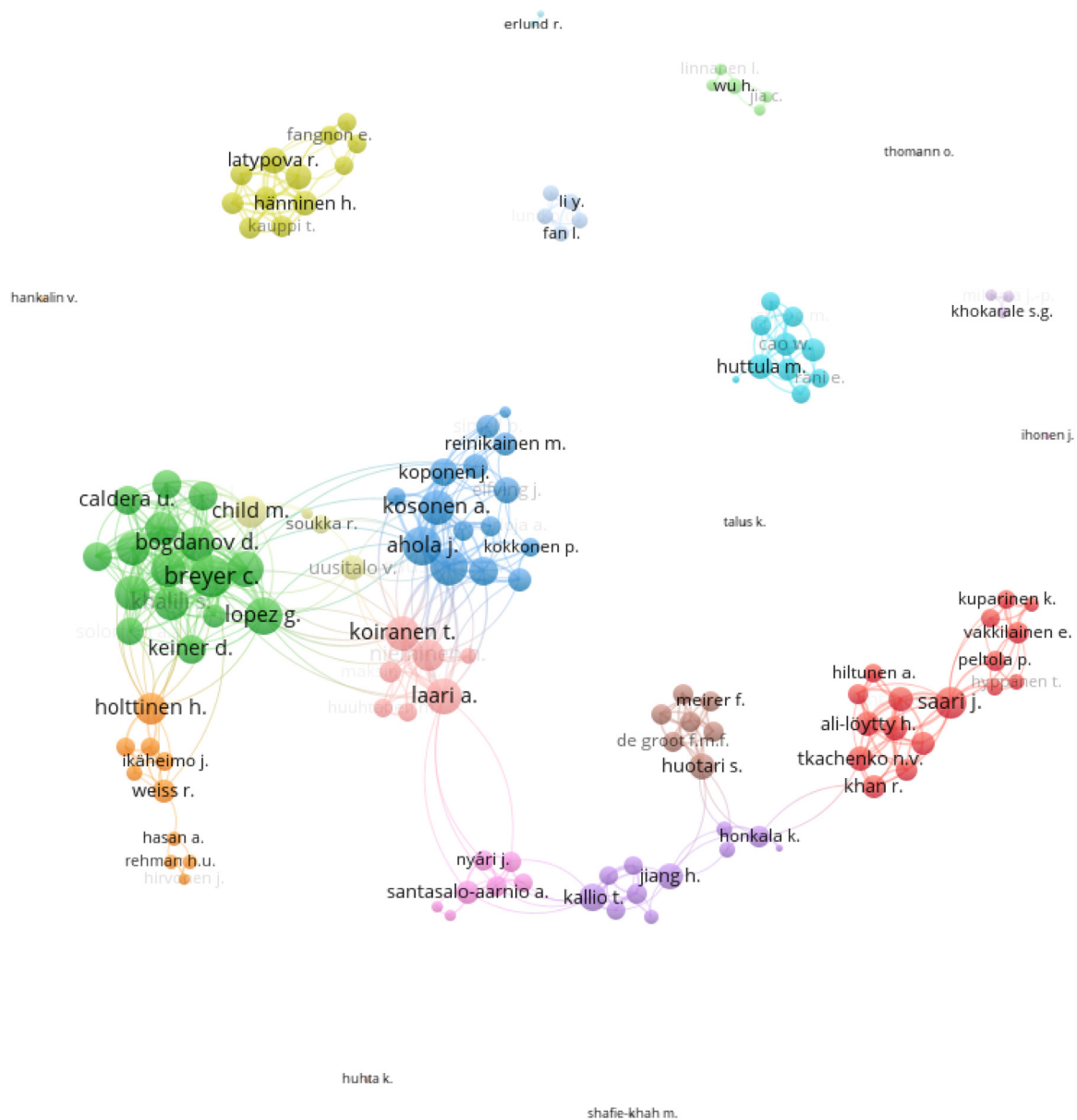


Fig. 7. Finnish PtX research teams VOSviewer analysis.

3.2.7. Normalized citation score

The Normalized Citation Score (NCS) analysis was conducted according to the publication date of the articles. The articles were allocated in the following groups: 2017–2018, 2019–2020 and 2021–2023. After this, the number of citations were averaged for each group (see Table 9). Last, the normalized citations score was done by dividing the total number of citations by the group average citation. Appendix 12 illustrates the top 20 articles by their normalized citation score.

An analysis was then conducted to investigate how the NCS was distributed by the year of publications and how authors and research groups, type of research and topics were represented in the 20 highest NCS score articles. The annual analysis illustrated that the yearly distribution of citations in different groups seemed to distribute evenly. However, the group 2021–2023 is not fully comparable because the number of citations will most likely increase the most in near future. The author analysis illustrated that most represented authors in the top 20 articles by NCS were Breyer C. ($n = 11$), Bogdanov D. ($n = 7$), Gulagi A. ($n = 5$), Fasihi M. ($n = 5$), Aghahosseini A. ($n = 4$) and Khalili S. ($n = 4$), who were involved in four articles or more. Moreover, the

analysis illustrated that research group one (Breyer et al.) was involved in ten out of twenty highest ranking articles by NCS. By research type, the analysis revealed that majority of the articles (18/20) were original research papers. Most topics in the top 20 articles were linked to the Power-to-X Economy, more precisely to energy system the energy system, and climate change topics. Other topics in the top 20 list were linked to electrolysis, e-ammonia, and CO₂ capture (DAC and industry). In addition, the analysis shows that here is only one raw material and material related article within the top 20 list by NSC.

The key takeaways from this section:

- Original research articles have higher NCS in the top 20 articles than review articles on this dataset.
- Energy system and climate change topics are most represented in top 20 NCS articles in Finnish PtX research.
- Other impactful topics by top 20 NCS articles are renewable hydrogen production linked to electrolysis, PtX technologies linked to e-ammonia, and CO₂ capture linked to DAC and industry sources.

Table 7
Top research teams in Finnish Power-to-X research.

Research group/dataset	Organization	Articles	Citations	AVG/Article
Breyer C. et al.	LUT University	48	2535	52.81
Laasonen K. et al.	Aalto University	11	529	48.09
Kiviluoma J. et al.	VTT	12	455	37.92
Ahola J. et al.	LUT University	21	403	19.19
Koiranen T. et al.	LUT University	17	200	11.76
Simell P. et al.	VTT	5	199	39.80
Lahtonen K. et al.	Tampere University	11	184	16.73
Huttula M. et al.	University of Oulu	9	143	15.89
Vakkilainen E. et al.	LUT University	11	109	9.91
Lund P.D. et al.	Aalto University	6	105	17.50
Sirén K. et al.	Aalto University	4	97	24.25
Santasalo-Aarnio A. et al.	Aalto University	8	86	10.75
Honkala K. et al.	University of Jyväskylä	7	86	12.29
Hänninen H. et al.	Aalto University	7	56	8.00
Huotari S. & Honkanen A-P.	University of Helsinki	3	55	18.33
Horttanainen M. et al.	LUT University	3	45	15.00
Zevehoven R. & Erlund R.	Åbo Akademi University	3	38	12.67
Huhtanen M. & Vuokila A.	University of Oulu	3	24	8.00
Kömi J. et al.,	University of Oulu	5	17	3.40
Linnanen L. et al.	LUT University	3	15	5.00

Table 8
Collaboration between top research groups.

RG1 Breyer et al., LUT UNI	RG4 Kiviluoma et al., VTT
RG16 Honkala et al., UNI Jyväskylä	RG5 Lahtonen et al. Tampere UNI
RG8 Koiranen et al., LUT UNI	RG12 Santasalo-Aarnio et al., Aalto UNI
RG7 Huotari & Honkanen., UNI Helsinki	RG16 Honkala et al., UNI Jyväskylä
RG3 Laasonen et al., Aalto UNI	RG12 Santasalo-Aarnio et al., Aalto UNI
RG3 Laasonen et al., Aalto UNI	RG16 Honkala et al., UNI Jyväskylä

Table 9
Averaged citations of the groups.

	2017–2018	2019–2020	2021–2023
Citations	2400	2920	2411
Number of articles	59	90	185
Average citation	40.68	32.44	13.03

3.2.8. Impact analysis

3.2.8.1. Topic impact analysis. The analysis of most impactful topics in Finnish PtX research started with analysis on the average of normalized NCS in category A (see Table 10). Here, the findings illustrate that the most impactful topics were CO₂ capture, PtX technologies and Power-to-X Economy. Carbon dioxide, renewable hydrogen production, appli-

cations, other by-products in hydrogen production and hydrogen had all average score less than one.

The analysis in the B and C categories (Table 10) included topics with at least 100 citations and five findings and average NCS of one. Here, the findings indicate that CO₂ capture impact was linked to industry-based carbon dioxide capture, faintly reflecting biogenic CO₂ capture. DAC had a higher average NCS than industry-based but had limited number of findings. In addition, our analysis illustrated that raw material and material related research linked to CO₂ capture represented a high average (2.03) NCS. The second most impactful topic in the category A was PtX technologies. Here, the analysis showed that Power-to-Gas (PtG), linked especially to the e-ammonia indicated high average NCS. In the PtX economy topic, the analysis revealed that environmental and sustainability aspects, especially in climate change are the most impactful topics by average NSC. Energy system integration and concepts represented also impactful topic, especially linked to the energy system research. The fourth most impactful category A, carbon dioxide, stood out by average NCS linked to its utilization. Renewable hydrogen production was the fifth most impactful topic by category A. Here, the most impactful topics in further categories link to electrolysis and photocatalysis. Moreover, the analysis illustrated that material and raw material research linked to renewable hydrogen production had higher than average NCS, especially linked to electrolysis (1.84). The research on applications was linked to the sustainable traffic system, when observing this through the average NCS, but with limited number of findings or

Table 10
Most impactful topics by Normalized Citation Score.

A-CATEGORY	AVG NCS	B-CATEGORY	# FINDINGS	AVG NCS	C-CATEGORY	# FINDINGS	AVG NCS
CO ₂ capture	1.19	(CO ₂ capture) DAC	6	4.48			
		(CO ₂ capture) Industry	31	1.13	(CO ₂ capture industry) bio	6	2.47
PtX technologies	1.18	(PtX) PtG	16	1.80	(PtG) e-ammonia	7	2.84
Power-to-X Economy	1.15	(Power-to-X Economy) Energy system integration and concepts	87	1.22	(Energy system integration and concepts) energy system	53	1.61
		(Power-to-X Economy) Environmental and sustainability aspects	37	1.66	(Environmental and sustainability aspects) climate change	24	2.32
Carbon dioxide	0.93	(Carbon dioxide) Utilization	21	1.19			
Renewable H ₂ production	0.92	(Renewable hydrogen production) Electrolysis	29	1.01			
		(Renewable hydrogen production) Photocatalysis	18	1.00			
Applications	0.79	(Applications) Sustainable traffic system	6	1.34			
Other	0.69	(Other) Heat	21	0.69	(Heat) utilization	9	1.08
Hydrogen	0.48						

Table 11
Impact analysis of collaboration.

	Number of articles	Total amount of citations	AVG citations	AVG NCS
Base set – all articles	334	7731	23.15	1.01
Domestic articles – no collaboration at all	145	3965	27.34	1.03
Only domestic collaboration – no industry collaboration	42	648	15.43	0.56
International collaboration – no filter	124	2660	21.45	1.15
Only international collaboration – no domestic collaboration	111	2354	21.12	1.19
Industry collaboration	40	911	22.78	1.18

representation in other topics. The impact of other by-products in hydrogen production was solely related to heat utilization but had a very limited number of findings. The impact on the hydrogen research was not highlighted in the impact analysis. Legislation and policy related findings did not indicate impact on the studies on the dataset, because the average NCS was not higher than two in any of the cases.

The key takeaways from this section:

- The most impactful topics emerge from CO₂ capture, PtX technologies, and Power-to-X Economy as follows:
 - PtG – e-ammonia
 - Power-to-X Economy – climate change and energy system.
 - Carbon dioxide – utilization
 - CO₂ capture – industry and biogenic CO₂.
 - Renewable hydrogen production – electrolysis and photocatalysis.
- Material and raw material research indicated high impact by NCS in CO₂ capture and renewable hydrogen production linked to electrolysis.

3.2.8.2. Collaboration impact. The impact of domestic, international, and industry-related collaboration is illustrated in Table 11. Analysis on domestic collaboration illustrates that domestic articles with no collaboration at all have the highest citation average. Moreover, it was found that articles done in sole collaboration with domestic organizations have the lowest citation and NCS average. However, there were only 42 articles that have been done in sole collaboration with domestic organizations. Thus, this is not comparable with other categories and it is not possible to draw conclusions based on this analysis.

The analysis of international collaboration highlights that it increases the impact of the Finnish research. More specifically, the average NCS of international collaboration, both general level (1.15) and in articles with only international collaboration (1.19), have higher average NCS than domestic articles. The industry collaboration also increases the impact of the research articles, compared with domestic articles. However, there are only 40 articles that have been done in industry collaboration, so this finding should be treated with caution.

The key takeaways from this section:

- International collaboration seems to increase the impact of Finnish PtX research, when measured through NCS.
- Slight indication that industry collaboration increases the impact of Finnish research by NCS, but this insight should be treated with caution as there is a limited number of these articles.

3.3. Survey findings

3.3.1. Research groups and teams in universities

The survey analysis started by mapping the research groups and teams across universities associated with PtX research in Finland. Here, the findings indicate that many of the institutions are still in the process of creating research groups and strategies around the PtX topic. This challenge arises due to the interdisciplinary nature of PtX research, necessitating collaboration between faculties and departments. Classifications of the faculties and departments were based on the Statistics Finland's Science 2010 Classification [50].

The findings indicate that the Aalto University, LUT University, Tampere University, University of Eastern Finland, University of Oulu, University of Turku, University of Vaasa, and Åbo Akademi University demonstrated a notable focus on engineering and technology faculties with emphasis on disciplines such as Chemical Engineering, Environmental Engineering, Materials Engineering, Mechanical Engineering, and Electronic, Automation, and Communications Engineering. The research activities of the VTT Technical Research Centre of Finland and Natural Resources Institute Finland, being research institutes, are concentrated on the realm of Engineering and Technology. In addition, the findings indicated a growing popularity of PtX research within the Social Sciences faculties at the LUT University, Tampere University, University of Eastern Finland, and University of Vaasa. The growing interest in this area, particularly in Business and Management studies and Law, underlines the interdisciplinary and multifaceted nature of PtX research. The involvement of natural sciences, such as Chemical Sciences, Physical Sciences, Computer and Information Sciences and Geography and Environmental Sciences, were highlighted in the University of Helsinki and University of Jyväskylä. The most interdisciplinary approach on Power-to-X Economy was exhibited by the Tampere University.

In summary, the faculties most involved in PtX research across Finnish universities predominantly falls under engineering and technology 80.2%, followed by natural sciences 13.8% and social sciences 6.0%. Further examination of the engineering and technology sector illustrates a diverse range of detailed research pursuits. Notably, 31% of the studies focus on chemical engineering, 24.1% on environmental engineering, and 15.5% on material engineering. In the natural sciences, the majority of faculty research is concentrated in chemical sciences (73.3%) and physical sciences (16.7%), with additional contributions from computer and information sciences and geography and environmental sciences. In addition, social science faculties are primarily divided among business and management (46.5%), law (38.5%), and political science (15.4%). These findings underscore the interdisciplinary nature of PtX research, with implications for policy and industry stakeholders.

The key takeaways from this section:

- The survey results underscore the interdisciplinary nature of PtX research, which requires collaboration between faculties and departments to create comprehensive research strategies and groups.
- Engineering and technology faculties show the highest concentration of PtX research, with a notable focus on disciplines such as chemical engineering, environmental engineering, and material engineering.
- Social sciences faculties also contribute to PtX research, with a growing interest observed in areas like business and management, law, and political science.

3.3.2. Research focus of the universities and organizations

The research focus of universities and organizations was classified according to a created framework. The most frequently addressed area in category A was renewable hydrogen production, representing 26.7% of the total category. This was followed by Power-to-X Economy (23.5%), hydrogen (17.3%), applications (14.7%), carbon Dioxide and CO₂ capture (11.4%) collectively, PtX technologies (4.9%), and other hydrogen production by-products, which accounted for only 1.6% of the total.

The most popular category A, renewable hydrogen production, was distributed in category B so that electrolysis was the most popular 35.4% share, followed by photocatalysis (28%), thermo-catalytic decomposition (9.8%), and power electronics and photosynthesis both had (1.2%) share. For 23.2% of the renewable hydrogen production there were no B category defined. Renewable hydrogen production did not have any representation in category C.

The second most popular category A was Power-to-X Economy, which was split into a category B with the following energy system integration and concept taking an 83.8% share and environmental and sustainability aspects accounting for the remaining 16.2%. The category B energy system integration and concepts was further divided into the category C, with the energy system holding a 49.1% share, value chain (21.1%), life cycle assessment and information technology each holding a 5.3% share, and machine learning/AI (3.5%). The energy system also included category D items such as optimization/flexibility (7.1%), off-grid solutions (10.3%), sector coupling (3.6%), although their majority were not categorized (85.7%). The environmental and sustainability aspect was divided into social acceptance (36.4%), biodiversity (18.2%), circular economy (18.2%), climate change (18.2%), and land use (9.1%).

The third most popular category A was hydrogen. This category has a wide range of categories of B, with utilization being the most mentioned (39.6%). Utilization of hydrogen also included categories C such as fuel cells (23.8%), but the majority of the category did not have any category C. Embrittlement was mentioned (20.8%), and storage and distribution of hydrogen had a combined 18.9% share. There was also one categorization of C about hydrogen pipelines (10%), while the others were blank. Compression of hydrogen had a share of 7.5% and the rest were not categorized (13.2%).

When looking into the applications, the process industry (33.3%) and sustainable steel (31.1%) are the topics getting the most focus. Sustainable traffic systems come in third, and they are divided into marine (72.7%), aviation (18.2), and mobile work machines/heavy machinery (9.1%). Pulp and paper industry has an 8.9% share, and sustainable metals were distinguished from sustainable steel and had a 2.2% share.

Carbon dioxide and CO₂ capture together constitute 11.4% of the total research focus. Of this, 7.5% is attributed to the carbon dioxide itself, while CO₂ Capture accounts for 3.9%. CO₂ Capture research is divided evenly between DAC and industrial CO₂ Capture. The focus of carbon dioxide research is mostly on utilization (73.9%), with some focusing on storing the carbon dioxide (4.3%).

The focus of PtX technologies research resides in Power-to-Liquids (PtL), specifically geared towards e-methanol and its derivative outputs, comprising a commanding 33.3% share of total PtX research efforts. Power-to-Material (PtM) research constitutes a significant 20% portion of PtX endeavors, while Power-to-Gas (PtG), including e-methane, is subject to a more modest 6.7% share. Other PtX research categories, lacking distinct categories in B. Other by-products in the hydrogen production together represent 1.6% of the research, with this modest percentage divided equally between research on solid carbon and the utilization of excess heat in hydrogen production.

The analysis also revealed the distribution of research in the terms of the share across material and raw material, legislation and quality and safety research. Legislation and policy accounted for approximately 5.3% of the research, while quality and safety constituted around 2.1% of the research. The largest portion, comprising roughly 35.6% of the research, focused on Raw Materials and Materials.

The key takeaways from this section:

- A strong emphasis on renewable hydrogen production and Power-to-X Economy, reflecting the transitioning to renewable energy sources and understanding the new energy system integration and concepts. Additionally, the inclusion of environmental and sustainability aspects in Power-to-X Economy research underscores the recognition

of the environmental impact and sustainability considerations in energy research.

- The focus on utilization in carbon dioxide research and the emphasis on Power-to-Liquids in PtX technologies research highlight the critical role of technological innovation in addressing the challenges of climate change and transitioning to a more sustainable energy future.
- Significant attention to raw material and material research.
- Overall, the findings from this analysis give complementary input to bibliometric study to better understand about the research priorities, which can guide future research agendas and policy initiatives in the field.

3.3.3. Laboratory and testing infrastructure related to Power-to-X research in Finnish universities

In the survey, the aim was to also ascertain the testing infrastructure related to Power-to-X Economy from Finnish universities. Universities faced challenges articulating infrastructure, as in many cases, the same testing infrastructure is utilized for research beyond PtX research.

3.3.3.1. Open access. The University of Oulu provides open access research infrastructures for scientific exploration. These include the large-scale MAX IV Laboratory Synchrotron Radiation Facility, offering X-ray-based characterization methods, and the Centre for Materials Analysis, equipped with various materials characterization instruments such as SEM, TEM, and XPS. Researchers can utilize these facilities for materials analysis at elevated temperatures and pressures, enabling the in-depth studies of chemical reactions.

At Aalto University, two key research infrastructures are available for open access, facilitating scientific exploration and collaboration. Both are operated together with the VTT. OTANANO is Finland's national research infrastructure dedicated to micro-, nano-, and quantum technologies. It comprises the Low Temperature Laboratory, Micronova, and the Nanomicroscopy Center. These facilities offer comprehensive micro- and nanofabrication capabilities, high-resolution imaging, and state-of-the-art experimental setups. RAMI is a national research infrastructure focused on raw materials research, aiming to enhance Finland's expertise in natural and inorganic material research for the circular economy. Aalto University contributes by processing, characterizing, and recycling primary and secondary raw materials, as well as exploring new material replacements for sustainable technologies.

The Power to X Laboratory at LUT University houses a variety of equipment and simulation tools focused on separation science and PtX technologies. The infrastructure supports research in carbon capture, utilization, and conversion processes. LUT University has also received funding to establish of the new PtX research laboratory at the Lappeenranta campus. The laboratory's focus will center on exploring synthetic electrical fuels and chemicals derived from water and carbon dioxide, poised to supplant conventional fossil fuels. The completion of the new laboratory is scheduled for 2025.

The University of Turku's Material Research Infrastructure (MARI) is a comprehensive university-level facility hosting various analytical instruments relevant to materials research. MARI's objective is to enhance materials research and technology in the Turku area, functioning as a collaborative platform to strengthen both teaching and scientific excellence. Open access is granted to researchers, providing opportunities for interdisciplinary studies, materials analysis, characterization, and aging investigations. The Turku Centre for Chemical and Molecular Analytics (CCMA) is a shared facility between the Department of Chemistry at the University of Turku and the Department of Science at Åbo Akademi University. This university-level infrastructure boasts five modern NMR instruments and a sophisticated LC-MS/MS mass spectrometer, establishing itself as one of the best-equipped educational NMR/MS facilities in Finland. Furthermore, the IT Center for Science Ltd (CSC) plays a pivotal role in Finland's scientific ecosystem, offering access to national computing resources, including multiple supercomputers.

3.3.3.2. Restricted access. In addition to the open access facilities, the University of Oulu hosts several research infrastructures with restricted access. These include specialized setups like the Tuning-fork Test setup for studying hydrogen embrittlement in steels and the Electric Arc Furnace Simulator for research on fossil-free steel making. These infrastructures are typically reserved for specific research groups or projects due to their specialized nature. Furthermore, the university provides access to membrane units, catalytic reactors, and electron/ion spectroscopy setups, all restricted to specific research needs. These infrastructures support studies ranging from gas-phase reactions to hydrogenation processes. Access to these restricted facilities is subject to approval and may require user fees to cover operation and maintenance costs.

Jyväskylä University's Department of Chemistry hosts several specialized research infrastructures catering to diverse scientific needs, particularly in the fields of materials science, spectroscopy, and microscopy.

Luke (Natural Resources Institute Finland) operates several biogas research infrastructures aimed at advancing the understanding and application of biogas production technologies. These infrastructures, ranging from laboratory-scale reactors to pilot and farm-scale biogas plants, facilitate research in biogas production and related processes. Access to these biogas research infrastructures is restricted and managed by Luke.

LUT University has also various modeling and simulation tools in use for PtX research. These models are so far available for the LUT University faculties, but several are planned for open source release in the foreseeable future.

Tampere University's Energy and Biorefining facilities focus on investigating the thermal conversion of fuels and feed stocks, including processes such as pyrolysis, gasification, combustion, liquefaction, catalytic pyrolysis, chemical looping, hydrothermal processing, and aqueous phase reforming. These techniques enable the conversion of various biomass and waste materials into high-value energy products. Operating at the university level, these facilities provide restricted access to researchers, offering essential infrastructure for studying sustainable energy production and refining. The Tampere Microscopy Center (TMC) is a university-level infrastructure dedicated to materials science and environmental engineering research. The Tampere Center for Scientific Computing (TCSC) operates the research infrastructure for computational science at Tampere University. As part of the Finnish Computing Competence Infrastructure (FCCI), TCSC contributes to advancing computational research and modeling in materials science and environmental engineering. Access to TCSC is restricted to university researchers.

The University of Vaasa's VEBIC (Vaasa Energy Business Innovation Centre) laboratories are currently developing a hydrogen fuel line. Additionally, they have a medium-speed engine designed for marine and power plant applications. Another ongoing project involves the construction of a single-cylinder research engine infrastructure. These facilities focus on various aspects of power generation with hydrogen, including peak shaving, combustion development, model-based powertrain design, powertrain control development, and fuel property characterization, and material interaction. The infrastructure is large-scale and restricted in access, with fees applicable. The technology readiness level (TRL) is estimated to be between 4 and 5, indicating that the technologies are in the stage of validation in a laboratory environment and are ready for further development towards industrial application.

The key takeaways from this section:

- **Challenges in Articulating Infrastructure:** Finnish universities faced difficulties in clearly defining testing infrastructure for PtX research, often due to shared usage with other research areas.
- **Open Access Infrastructures:** Universities like the University of Oulu, Aalto University, and LUT University provide open access to state-of-the-art research facilities, supporting PtX research and fostering collaboration.
- **Restricted Access Infrastructures:** Specialized setups and facilities, such as those at University of Oulu, Jyväskylä University, and Luke,

offer restricted access to specific research needs, indicating a focus on specialized studies within PtX and related fields.

4. Discussion

4.1. Term analysis

During the bibliometric analysis, the authors noticed an inconsistency in terminology usage within the academic literature surrounding the PtX concept. It became apparent that terms such as "Power-to-X" itself, along with "PtX" and "P2X," were frequently used interchangeably to refer to the same idea: the conversion of power into more valuable products. The observed lack of consensus not only leads to fragmentation within the field, but also serves as a foundation for numerous derived alternative terms. For instance, the process of converting power to synthetic liquid hydrocarbons, sometimes referred to as "power-to-liquids", is referred to in the literature by at least three terms: "power-to-liquid", "PtL", and "P2L" in different studies [51–53]. This inconsistency not only limits the potential exposure of some articles that use less popular terms but also complicates the systematic analysis of the field. Therefore, to gain a better understanding about the evolving usage of these terms, this study systematically evaluated the frequency of PtX related terms in academic literature from 2010 to 2023.

The findings of the study strongly indicated that "Power-to-X" itself is the most popular term used in article metadata when discussing the PtX concept during the years immediately preceding this study (2020–2023). During the same periods the terms "PtX" and "P2X" were used scarcely. Additionally, when searching for these terms, many additional articles related to P2X receptor study, a field of medicine, or PTX (an abbreviation for Paclitaxel) chemotherapy agents, appeared alongside the PtX related articles. Therefore, for optimal exposure in the systematic analysis of the PtX, the utilization of the "Power-to-X" term exclusively in the article metadata (title, abstract and keywords) is of importance.

4.2. Bibliometric and survey-interview analysis

4.2.1. Research focus, impact and gaps

Based on the results of this study, the volume of Finnish PtX research exceeds the most published countries in hydrogen economy [54] and green hydrogen [55] related research. The leading focus of Finnish PtX research is in the Power-to-X Economy, especially linked to energy system research. In addition, environmental and sustainability aspects were in the focus, namely climate change, which was often research together with topics linked to energy system integration and concepts. Other popular topics were CO₂ capture linked to industry sources, renewable hydrogen production linked to electrolysis, PtX technologies linked to PtL (e-methanol and derived products), hydrogen (storage), carbon dioxide (utilization), and other by-products in renewable hydrogen production (heat).

The legislation and policy, and material and raw material research in Finnish PtX research were also visible. The material and raw material research was especially linked to renewable hydrogen production and CO₂ capture. Moreover, the findings indicate that research is established on legislation and policy, both of which is focused on general level topics of energy system integration and concepts and environmental and sustainability aspects. In addition, the survey and interview findings indicate that research is dominated by engineering departments and that the research focus is on renewable hydrogen production and Power-to-X Economy. Also, climate change and green transition are in the focus of different research organizations, which is linked on the technological innovations addressing these topics. Consequently, the analyzes seem to support each other, from which it can be concluded that the Finnish PtX research has a technological focus.

When compared with international level hydrogen economy research [54], the similarities on focus are on the renewable hydrogen production and storage and distribution of hydrogen. The difference is

that Finnish PtX focuses more on Power-to-X Economy, environmental and sustainability aspects, CO₂ capture, renewable hydrogen utilization, and PtX technologies. In addition, Finnish PtX economy research does not address the application of hydrogen or policy and safety, compared with international level hydrogen economy related research [54].

The impact of Finnish PtX research is very similar to its volume, as Power-to-X Economy is the most cited topic, linked to energy system research. Moreover, environment and sustainability aspects, namely climate change, has gained significant interest from academia, but this is probably not the focus of studies but more of a variable. Other clearly influential research topics are CO₂ capture linked to industry sources and biogenic CO₂ and DAC, renewable hydrogen production mostly linked to electrolysis and PtX technologies linked to PtG and e-ammonia. However, it is important to mention that there are several other influential topics in Finnish PtX research that are represented in their own sections. The NCS analysis yielded similar results but seem to highlight CO₂ capture and PtX technologies over the Power-to-X Economy. Moreover, the NCS analysis indicates that environmental and sustainability aspects, especially climate change, has more impact than energy system-oriented research.

This study illustrates that there are numerous research gaps in the field of Finnish PtX economy (see Appendix 2). These apply to both popular and less researched topics within the framework of this study that researchers need to investigate in order to promote advancements in the field of PtX. In the first case, the PtX economy lacks research in both of its aspects. Firstly, energy system integration and concepts have research gaps in value chain and business model research, life cycle assessment, security of supply, infrastructure, hydrogen valleys, machine learning and AI, and information technology. Secondly, environmental and sustainability aspects represent research gaps in wealth creation, circular economy, biodiversity, land use and sustainability. The research gaps in other popular topics are as follows. The renewable hydrogen production lacks research in methane pyrolysis, photosynthesis, power electronics, alkaline, and PEM and only a little research in photovoltaic, solid oxide electrolysis cell, and AEM. CO₂ capture topic represents minimal research in DAC and it would also benefit from more research on industry sources, e.g., in biogenic sources which has a little research but high impact. Hydrogen utilization research is focused only on heat, and consequently there are research gaps in electricity, fuel cells, gas turbines, and combustion engines. In hydrogen distribution, there is no research at all in compression, pipelines or maritime. In hydrogen storage, there is no research in the gas, liquid and metal form of storage and in compression and only a little research in hydrogen embrittlement. In the PtX technologies topic the analysis illustrates limited research in PtG in e-methane, PtL in e-fuels and e-ethanol, and PtM in e-food, e-cement, and e-plastics.

The research gaps in less popular topics are as follows. The carbon dioxide topic illustrates limited research on its storage and distribution. Application topic show research gaps in several fields, that are the sustainable traffic system (aviation and mobile working machines), sustainable metals (hydrogen reduction and hydrogen in process heating), chemical industry (pharmaceutical and plastics), and in agriculture, cement industry, manufacturing industry, process industry, paper and pulp industry, power and heat industry. Other by-products of hydrogen production topic lacks research on solid carbon, oxygen, and nitrogen.

The legislation, material and quality aspects represent significant research gaps in the Finnish PtX research. The material and raw material research could expand its focus from renewable hydrogen production, e.g., to CO₂ capture and hydrogen (embrittlement, distribution, and storage), which have currently limited research. Legislation and policy research has not addressed topics outside the Power-to-X Economy, which represents a major research gap. Moreover, quality and safety aspects seem to be missing in Finnish PtX research.

The survey and interview results indicate that research organizations are aiming to adjust their research focus in the future. Here, the findings

indicate that renewable hydrogen production, Power-to-X Economy and hydrogen will remain strong research interests. However, there seem to be an increasing research interest in renewable hydrogen production towards thermos-catalytic decomposition and other alternative hydrogen production methods. In Power-to-X economy, the future focus seems to include value chain (regarding energy system integration and concepts) and social acceptance, biodiversity, and circular economy (in environmental and sustainability aspects). In hydrogen research the future research orientation seems to show interest towards its utilization in fuel cells. Application research indicates radical change towards process industry and sustainable steel, as they represent 66% of the interest reported by the Finnish research organizations. The PtX technologies research seem to steer away from the PtG into PtL (linked to e-methanol and derived products) and into PtM.

4.2.2. Research organizations focus, collaboration and authors and research groups involved

The focus of Finnish research organization is divided into two groups: i) organizations with broad focus and ii) organizations with more specific focus on the topics in the framework of this study. Moreover, the survey and interview results indicated that some of the universities aim to expand their focus in the future. The number of departments involved might be one of the explanatory factors in the research focus of the organizations and thus it will be necessary to monitor does the increasing number of departments involved in the PtX research increase the focus of research organizations and tackle the above-mentioned research gaps.

The analysis of collaboration in Finnish research organizations reveals that most of the collaboration is done internationally. More precisely, the international collaboration takes place mostly within Europe, but China is the single most popular international collaboration destination. Otherwise, the analysis reveals that international collaboration is limited in other continents than Europe and Asia. In addition, it was discovered that international collaboration increases the impact of research, which corresponds to findings with other bibliometric analysis on green hydrogen research [55]. Moreover, the findings indicate that domestic collaboration was limited compared with international collaboration. This is noteworthy insight for Finnish research, because organizations have several overlaps and areas of specialization that could be utilized in PtX research. Also, the findings of the surveys and interviews indicate limited internal collaboration within Finnish research organizations, which is a key development area for the future. This insight is in line with other bibliometric studies, which highlight the need for interdisciplinary cooperation between different disciplines [54].

The findings also illustrated that there is a strong and growing correspondence of authors working on the PtX research. The most productive researchers associated in Finnish PtX research have published more articles compared with international bibliometric analysis on hydrogen economy and green hydrogen [54,55] and exceed the number of citations of top authors in green hydrogen research [55]. However, the most productive authors seem to represent a few specific organizations. Moreover, the analysis highlighted several active research groups within PtX research, but the majority of them were also linked on certain organizations. In addition, the analysis indicated that collaboration between the most productive research groups was limited. These insights are supported by the survey and interview findings, which indicate that there is limited collaboration between the research groups.

5. Conclusions

This study provides a comprehensive overview of Finnish Power-to-X Economy research from multiple perspectives. Overall, the results indicate that it withstands the international comparison by volume but shows also that there is still plenty to do in the future. Consequently, this study highlights the research strengths and gaps through which knowledge on a very broad topic can be promoted.

In addition, this study provides suggestions of how the quality of Power-to-X research can be improved. First, the results indicate that it would be necessary to increase the domestic collaboration between research organizations as this is quite limited. In this connection, it is also necessary to mention that one way of doing this could be to open the collaboration between the established research groups identified in this study. Second, interdisciplinary research is warranted. This is a key observation in both bibliometric research and interviews and survey, through which significant advancements in research can be achieved. Our analysis offers valuable insights into the ongoing developments and the early-stage establishment of Power-to-X research groups across Finnish universities. The interdisciplinary nature of Power-to-X research is underscored by the collaborative efforts between faculties and departments, as reflected in the diverse engagement in engineering, social sciences, and natural sciences. This comprehensive approach not only exemplifies the transformative potential of Power-to-X research but also underscores its significance in addressing pressing global challenges.

This study also has significant international implications for the academic fields focusing on Power-to-X. First, the framework created in this study can help to map the different elements of Power-to-X research and hopefully future research will develop this further on the global scale. Second, this study encourages for similar overviews in other countries. More importantly, future research is needed to compare the results of different countries, which would enable a more comprehensive view on the state of Power-to-X research. Moreover, one could observe the features driving the development of Power-to-X research and how these differ between countries. This would also provide excellent conditions for improving international research cooperation and thereby increase the knowledge on Power-to-X and achievement of the global climate goals.

As a practical implication, it would be recommendable for researchers to link their Power-to-X studies clearly, e.g., in abstracts or keywords. This would improve the visibility and availability of Power-to-X research and thus enable efficient use in research, product development and investments. In addition, this study showed that research organizations have several areas of specialization in Power-to-X research that are also overlapping with other organizations. Moreover, the strengths can also be applied to address the research gaps or development areas between different organizations. Thus, organizations can utilize the findings of this study to promote the research collaboration and improve the impact and quality of Power-to-X research. It would be also useful to highlight if studies involve practical projects, experimental research, pilot plant activities or modelling, so that these could be more easily mapped and utilized.

For policymakers, the overall insight is that funding and other supportive actions should be directed towards the research gaps highlighted in this study. Research advancements on the Power-to-X Economy significantly support the green transition in Finland. This in turn enables a more educated population and the emergence of new innovations and businesses. Overall, if Finland achieves its climate objectives, it would be a forerunner in the green transition, and the lessons learned from it could be utilized on a global level. Decision makers could evaluate these topics and allocation of research funding through which are significant from the Finnish perspective and thus promote the Power-to-X Economy development nationally. Funding plays a significant role in the creation of new knowledge and supporting technology development in Power-to-X Economy, which in turn contributes to the advancement in green transition policies and initiatives. As a concluding remark, this study unlikely mapped out all the Power-to-X research conducted in Finland. Although a lot of effort was put into the framework of this study, this needs to be further developed in order to for this to cover all the aspects of Power-to-X research. In addition, the terminology of Power-to-X research is highly versatile and complicated and there are several overlaps of its abbreviations with other disciplines and subjects, which makes it extremely difficult to map the entire Power-to-X research conducted.

Thus, further research needs to be conducted to validate the framework and procedures of this article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Teemu Tuomisalo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eeva Lähdesmäki:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lisa Ulvila:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Christian Breyer:** Writing – review & editing, Validation, Supervision. **Siavash Khalili:** Writing – review & editing, Visualization. **Samuli Urpelainen:** Writing – review & editing, Formal analysis, Conceptualization. **Petteri Laaksonen:** Writing – review & editing, Supervision, Conceptualization.

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Supplementary materials

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